

SOUMYA SAMANTA

Backend Software Engineer — Java / Spring Boot

Kolkata, India | +91-7318939350 | soumyasamanta354@gmail.com | [LinkedIn](#) | [GitHub](#)

SUMMARY

Backend Software Engineer with 2+ years of experience building and optimizing Java/Spring Boot microservices for high-volume banking platforms (SBI YONO, West Bengal IPKS, Tamil Nadu IFSS). Skilled in REST API design, third-party service integration (NSDL, DigiLocker, CKYC, payment gateways), and SQL performance tuning, with measurable improvements to API latency and query execution time. Comfortable working across the stack when needed, with a primary focus on scalable, reliable backend systems.

WORK EXPERIENCE

Software Engineer | Tata Consultancy Services (TCS)

Aug 2024 - Present

- Developed and integrated OTP, SMS, email, and in-app push notification services for SBI YONO, strengthening notification delivery reliability across authentication and transaction workflows.
- Developed and integrated third-party REST APIs for banking workflows, standardizing request/response handling, validation, and error management to improve service integration efficiency.
- Enhanced backend services for West Bengal IPKS using Java, Spring Boot, JDBC, and SQL, reducing API response times by approximately 15% through backend and query-level optimizations.
- Integrated WhatsApp notification services into banking workflows, automating customer communication and reducing manual notification effort.
- Contributed to Tamil Nadu IFSS core banking modules, integrating NSDL, DigiLocker, CKYC, and payment gateway services to streamline KYC, onboarding, and payment workflows.
- Optimized SQL queries and database interactions, reducing query execution time by approximately 20% and improving performance and reliability of high-volume banking operations.
- Used AI-assisted tools (GitHub Copilot, Claude, ChatGPT) to accelerate prototyping, debugging, and SQL query optimization during feature development.

PROJECTS

Common Exam and Learning App (CELA) | React, Spring Boot, Microservices, MySQL

Jan 2025 - May 2025

- Built a multi-role platform for students, organizations, and admins to conduct exams, manage learning content, and track performance.
- Implemented role-based authentication (JWT/Spring Security) and real-time analytics dashboards for exam performance tracking.
- Designed a scalable microservices architecture with MySQL persistence, enabling independent deployment of exam, learning, and analytics modules.

WhatsApp Chat Analyser | Python, Pandas, Matplotlib, Streamlit, Regex

Jan 2024 - May 2024

- Built a web app that parses exported WhatsApp chat logs and generates statistics on message volume, active hours, most-used words/emojis, and per-participant activity trends.
- Used regex-based parsing to structure raw chat exports into a clean dataset, then processed it with Pandas for time-series and frequency analysis.
- Built interactive visualizations (timelines, heatmaps, word clouds) with Matplotlib and deployed the tool as a Streamlit web app for non-technical users.

SKILLS

Languages: Java, Python, SQL, C++, C, JavaScript

Backend: Spring Boot, Microservices, REST APIs, Apache Kafka, JUnit, Mockito, Swagger/OpenAPI

Databases: MySQL, PostgreSQL, Redis

Frontend (supporting): HTML, CSS, React

Tools & Practices: Git, GitHub, Docker, AWS (basics), VS Code, Eclipse, STS, Agile/Scrum, System Design, Design Patterns, OOP, Data Structures & Algorithms, Unit/Integration Testing, TDD, DBMS, Operating Systems, Computer Networks

CERTIFICATIONS

- Apache Kafka — Udemy
- Spring Boot — Udemy

EDUCATION

B.Tech, Computer Science and Engineering

2020 - 2024

Ramkrishna Mahato Government Engineering College, Purulia — CGPA: 9.12/10

ACHIEVEMENTS

- Solved 800+ problems on LeetCode, achieving a max rating of 1711+ (3-star).
- Solved 80+ problems in CodeChef contests, achieving a max rating of 1604+ (3-star).
- Active coding profiles: LeetCode, CodeChef, GeeksforGeeks, GitHub.